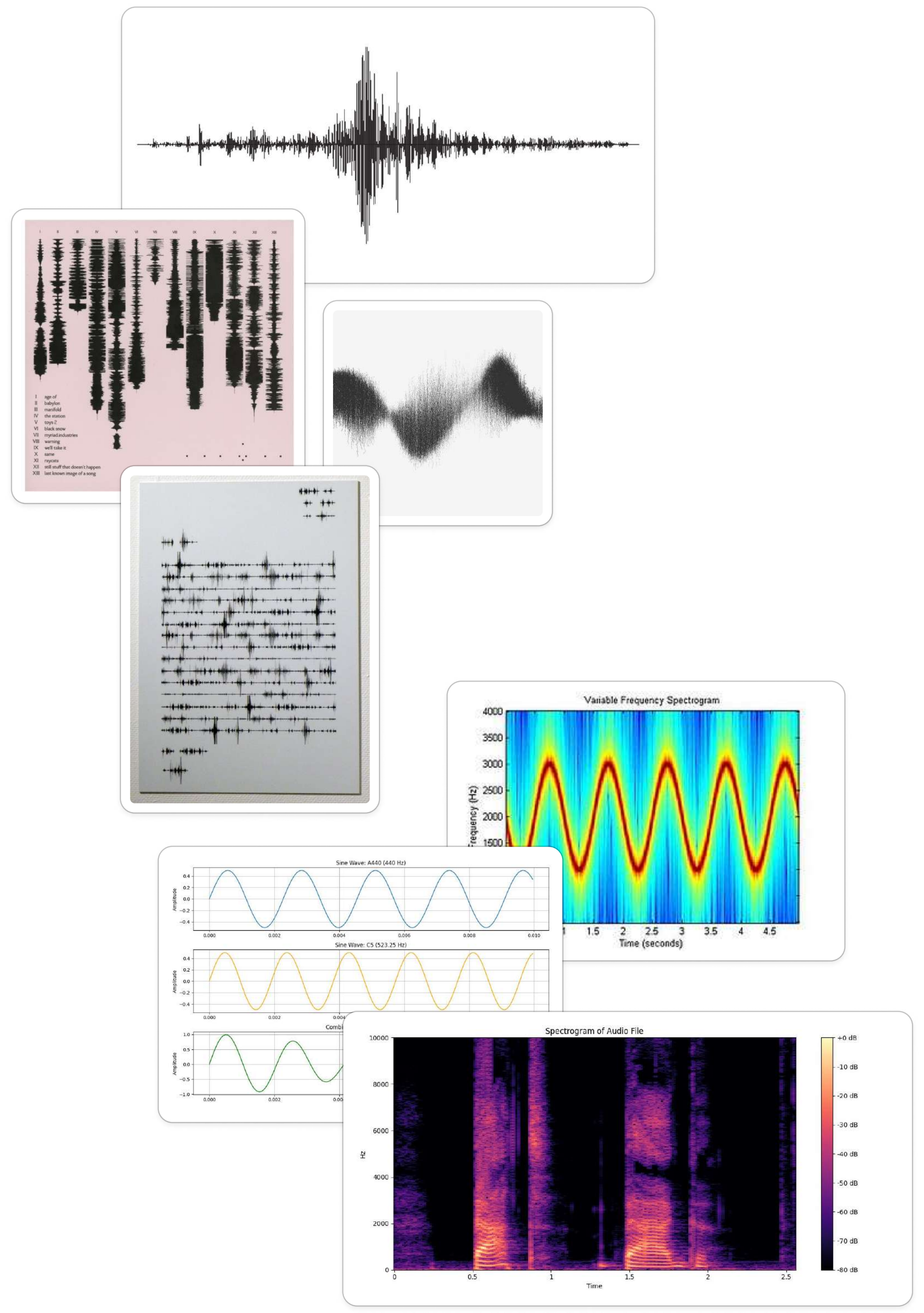
Section 1

ONE SONG → MANY WAYS OF REPRESENTING IT

How many different ways can the same song exist?

LOUD / QUIET / TIME / PAUSES

WHAT IS BEING SUNG / WHAT IT MEANS / WHAT IT FELT LIKE

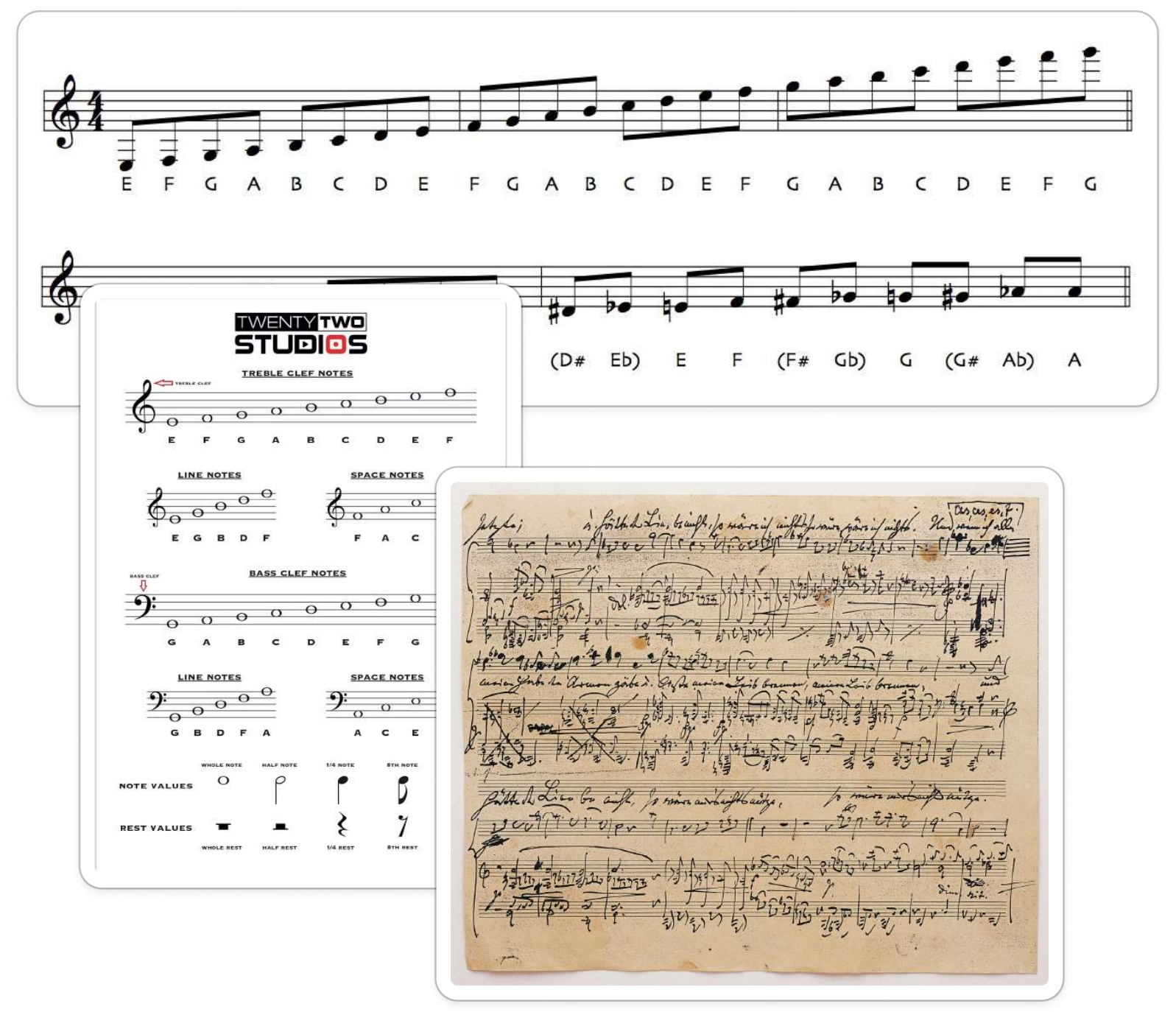


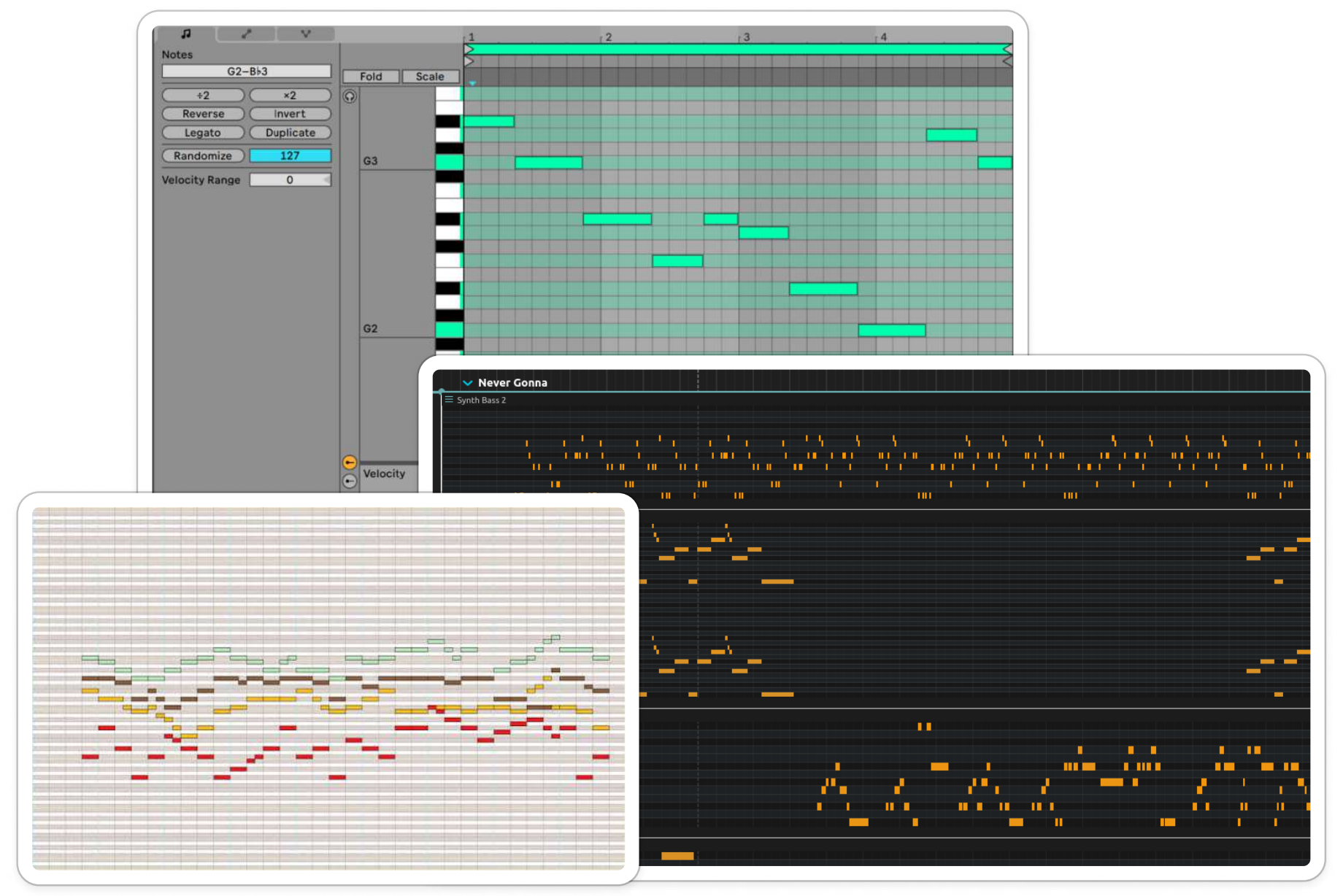
FREQUENCY / TIME / INTENSITY / SOUND TEXTURE

LYRICS / MEANING / CONTEXT / EXPERIENCE

NOTES / RHYTHM / TIMING / STRUCTURE

ACTUAL VOICE / CROWD / SPACE / EXPERIENCE



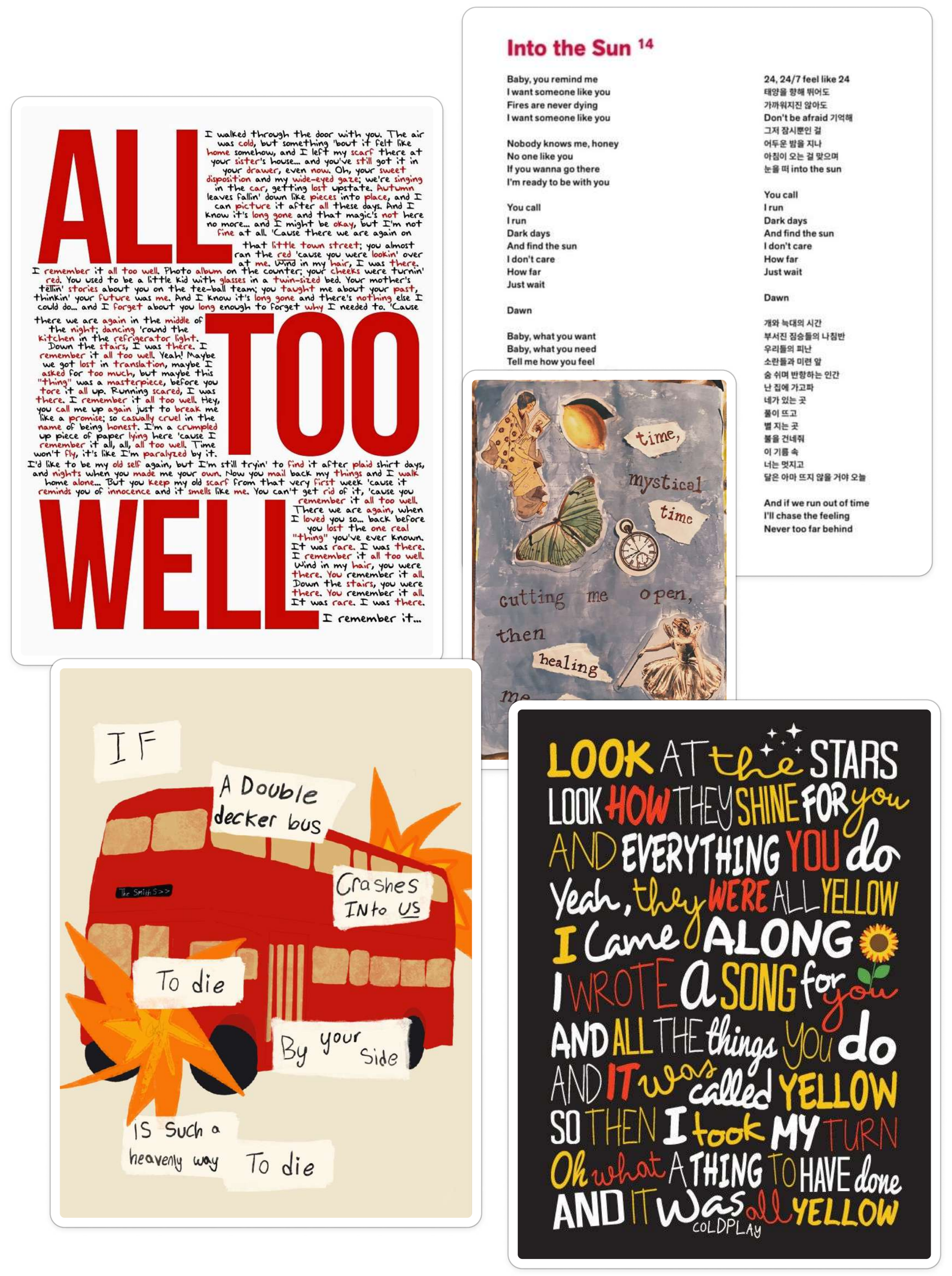


NOTES / TIME / DURATION / STRUCTURE

LYRICS / LIVE PERFORMANCE / CONTEXT / EXPERIENCE

WORDS / ORDER / LANGUAGE / MEANING

MELODY / VOICE / SOUND / PERFORMANCE


ALL TOO WELL
I walked through the door with you. The air was cold, but something 'bout it felt like home somehow, and I left my scarf there at your sister's house... and you've still got it in your drawer, even now. Oh, your sweet disposition and my wide-eyed gaze; we're singing in the car, getting lost upstate. Autumn leaves fallin' down like pieces into place, and I can picture it after all these days. And I know it's long gone and that magic's not here no more... and I might be okay, but I'm not fine at all. 'Cause there we are again on that little town street; you almost ran the red 'cause you were lookin' over at me. Wind in my hair, I was there. I remember it all too well. Photo album on the counter; your cheeks were turnin' red. You used to be a little kid with glasses in a twin-sized bed. Your mother's tellin' stories about you on the tee-ball team; you taught me about your past, thinkin' your future was me. And I know it's long gone and there's nothing else I could do... and I forget about you long enough to forget why I needed to. 'Cause there we are again in the middle of the night; dancing 'round the kitchen in the refrigerator light. Down the stairs, I was there. I remember it all too well. Yeah! Maybe we got lost in translation, maybe I asked for too much, but maybe this "thing" was a masterpiece, before you tore it all up. Running scared, I was there. I remember it all too well. Hey, you call me up again just to break me like a promise; so casually cruel in the name of being honest. I'm a crumpled up piece of paper lying here 'cause I remember it all, all, all too well. Time won't fly, it's like I'm paralyzed by it. I'd like to be my old self again, but I'm still tryin' to find it after plaid shirt days, and nights when you made me your own. Now you mail back my things and I walk home alone... But you keep my old scarf from that very first week 'cause it reminds you of innocence and it smells like me. You can't get rid of it, 'cause you remember it all too well. There we are again, when I loved you so... back before you lost the one real "thing" you've ever known. It was rare. I was there. I remember it all too well. Wind in my hair, you were there. You remember it all. Down the stairs, you were there. You remember it all. It was rare. I was there. I remember it...
time,
mystical
time
cutting me open,
then
healing
me
IF
A Double decker bus
The Smiths>>
Crashes INto US
To die
By your Side
IS Such a heavenly way
To die
LOOK AT the STARS
LOOK HOW THEY SHINE FOR you
AND EVERYTHING YOU do
Yeah, they WERE ALL YELLOW
I Came ALONG
I WROTE a SONG for you
AND ALL THE things YOU do
AND IT was called YELLOW
SO THEN I took MY TURN
Oh what A THING TO HAVE done
AND IT Was all YELLOW
COLDPLAY

Into the Sun [14]

Baby, you remind me
I want someone like you
Fires are never dying
I want someone like you

Nobody knows me, honey
No one like you
If you wanna go there
I'm ready to be with you

You call
I run
Dark days
And find the sun
I don't care
How far
Just wait

Dawn

Baby, what you want
Baby, what you need
Tell me how you feel

24, 24/7 feel like 24
태양을 향해 뛰어도
가까워지진 않아도
Don't be afraid 기억해
그저 잠시뿐인 걸
어두운 밤을 지나
아침이 오는 걸 맞으며
눈을 떠 into the sun

You call
I run
Dark days
And find the sun
I don't care
How far
Just wait

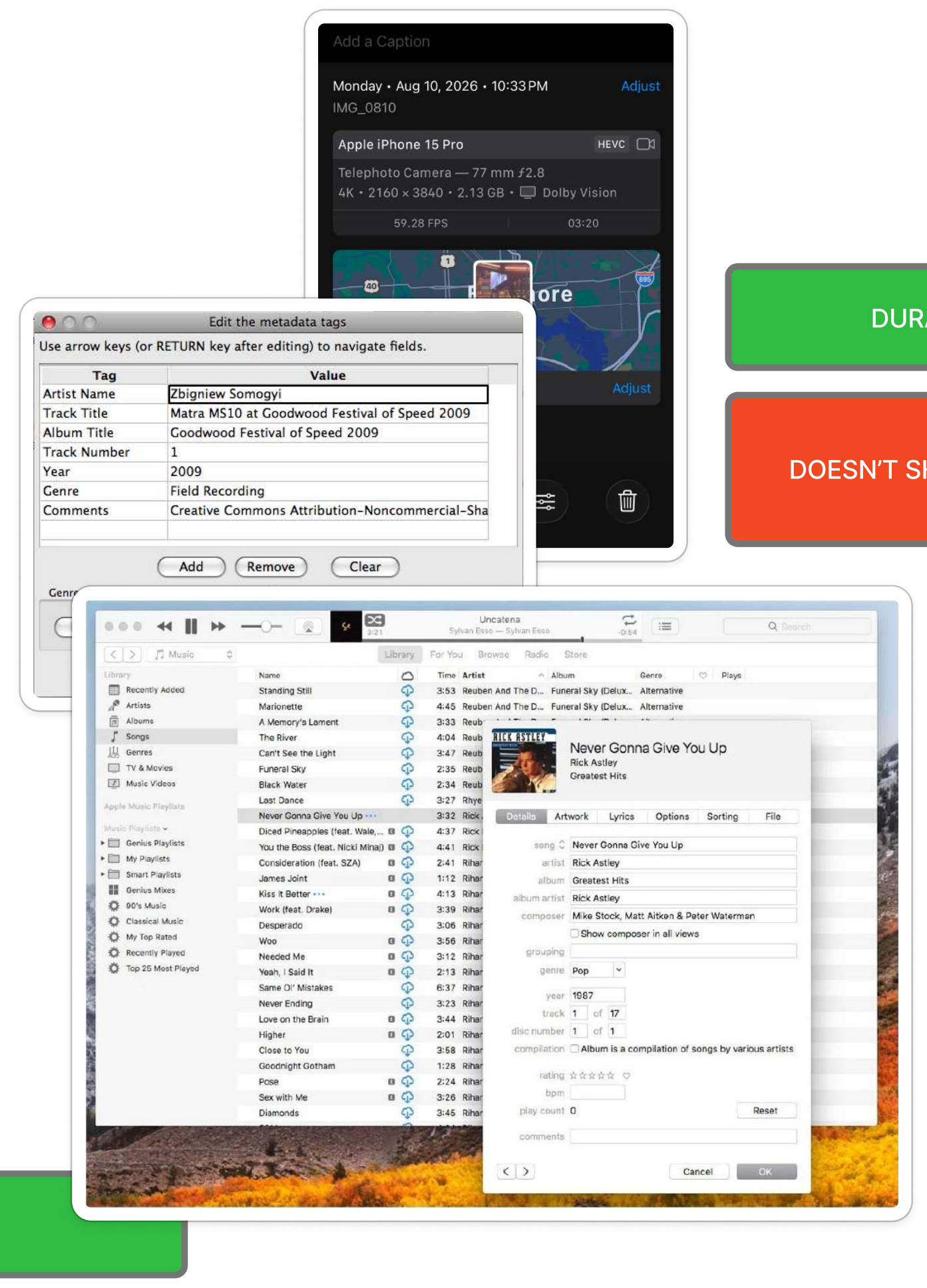
Dawn

개와 늑대의 시간
부서진 짐승들의 나침반
우리들의 피난
소란들과 미련 앞
숨 쉬며 반항하는 인간
난 집에 가고파
네가 있는 곳
풀이 뜨고
별 지는 곳
불을 건네줘
이 기름 속
너는 멋지고
달은 아마 뜨지 않을 거야 오늘

And if we run out of time
I'll chase the feeling
Never too far behind

SOUND / MOVEMENT / TIME / ONE VIEWPOINT

PHYSICAL SCALE / BODY / FULL SPACE / EVERYTHING OUTSIDE FRAME

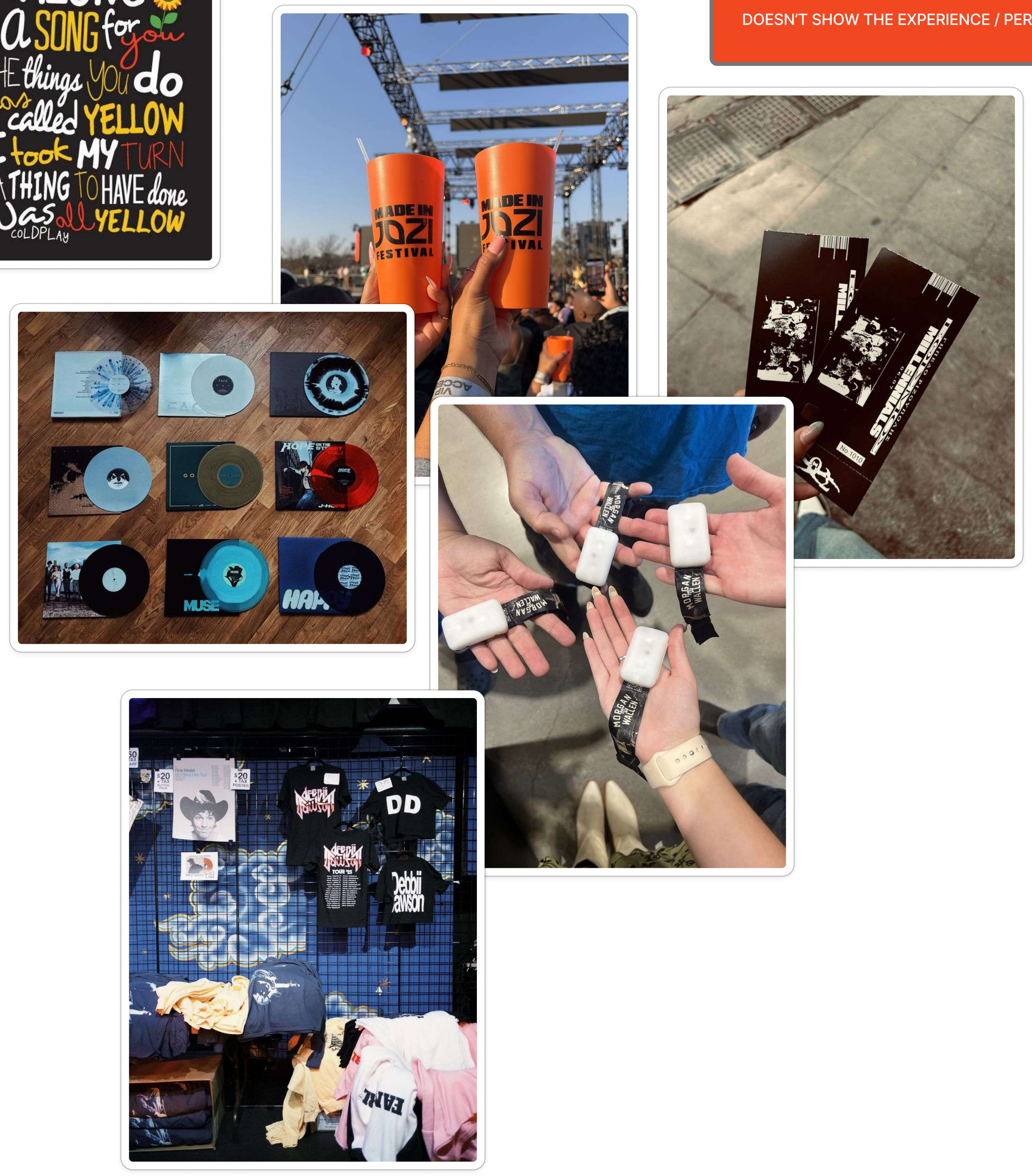


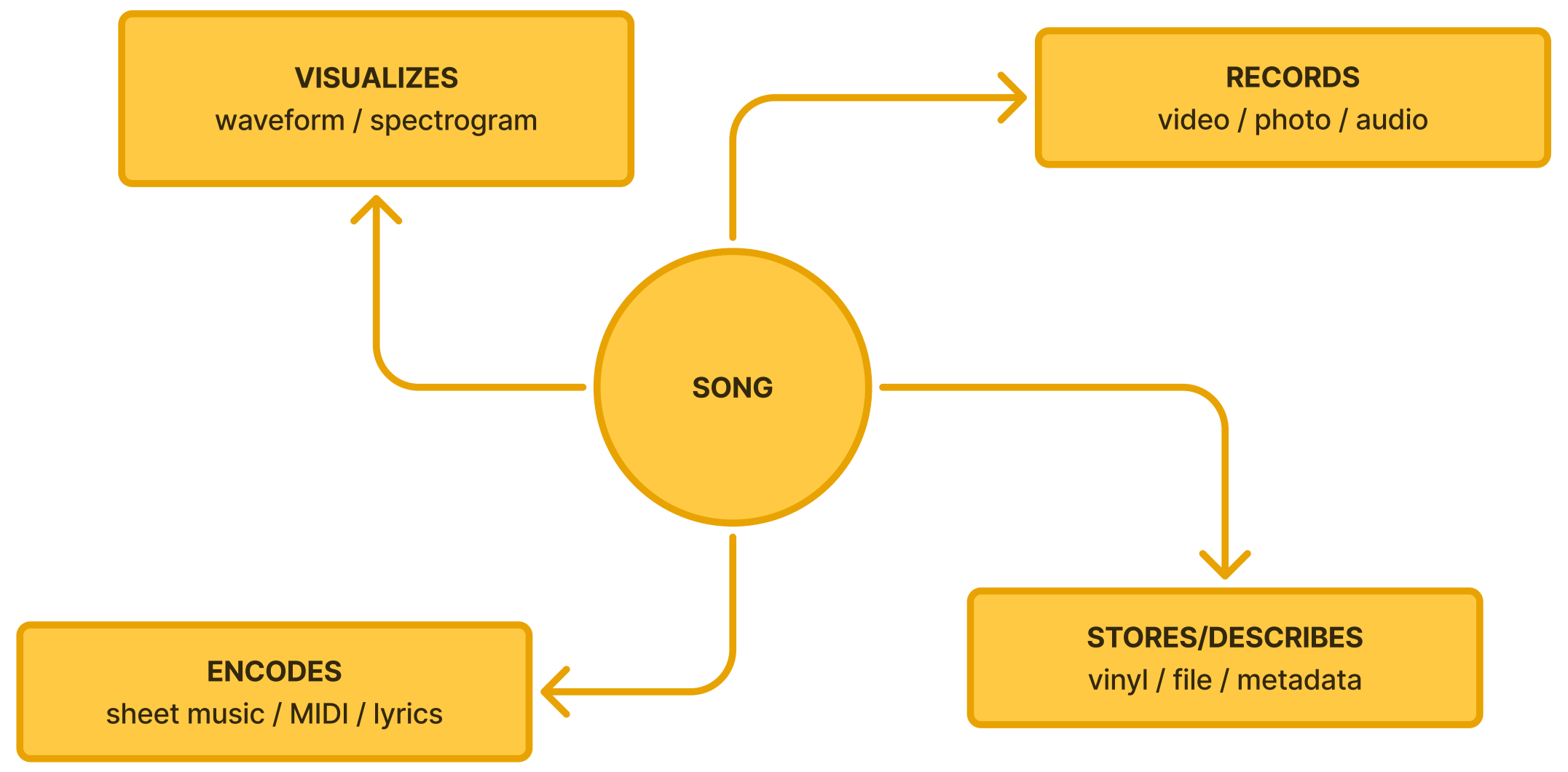
DURATION / FILE TYPE / DATE / SIZE / BPM / KEY

DOESN'T SHOW WHAT IT SOUNDS LIKE / MEANS / FEELS LIKE

STORES / CARRIES / RETRIEVES SOUND

DOESN'T SHOW THE EXPERIENCE / PERFORMANCE / CONTEXT





VISUALIZES
waveform / spectrogram

RECORDS
video / photo / audio

SONG

ENCODES
sheet music / MIDI / lyrics

STORES/DESCRIBES
vinyl / file / metadata

Section 2

What makes a response intelligent?

Is responding enough?

Or does intelligence require:

LISTENING → MEMORY → INTERPRETATION → CHOICE → RESPONSE?

INPUT → INTERPRETATION/RULE → RESPONSE

Where is the intelligence?

Prachee Shah

- MEMORY → How do we recognize a sound we've heard before?
- INHERITED INTELLIGENCE → What sound behaviors are learned vs somehow already available to us/species?
- TONE GENERATES → A tone itself can become a trigger.
- DIFFERENT SPECIES → Bird calls, whales, insects, etc. Sound as communication without human language.
- SPURRING PHENOMENA → One sound causes another behavior/ event.
- ABSENCE OF VISUAL STIMULI → How much intelligence/ communication can happen through sound alone?

TRIGGER → RESPONSE

Sound as trigger → intelligent response

VOICE → RESPONSE

CALL → RESPONSE

MUSIC CUE → LIGHTSTICK

SPORTS CHANT → CROWD

IMPROVISATION

AUTOTUNE

FOLEY

Everyone is following instructions together.

Collective Intelligence

Thousands of separate light sticks working together during a concert

Music is playing → there's a cue → lightsticks change color/pattern → the whole stadium creates one visual.

One lightstick by itself isn't really doing anything interesting. The pattern only happens when thousands of them are connected.

If everything is just following instructions, where is the intelligence?

Where is the intelligence?
- Is it in the lightstick?
- Is it in the program controlling them?
- Is it in whoever programmed the show?
- Or is the whole connected system the "smart" part?
- The crowd looks like it's reacting to the music, but is it actually reacting or just following instructions?

A fake concert crowd made of dots/ lights. Play sound or give it a cue → different areas light up. Then mess with the system by removing some lights or changing the input and see what happens.

I hear what you did and decide what I should do back.

PERSON A → SOUND → PERSON B INTERPRETS → RESPONSE → PERSON A...

One person makes a sound and another person responds to what they heard.
Could be musicians improvising, a singer and audience, chants at concerts/sports, even normal conversation.

Person A makes sound → Person B hears it → figures out what they heard → decides how to respond → makes another sound → Person A responds back.

What makes a sound a response instead of just another sound?

Where is the intelligence?
- Knowing when it's your turn
- Recognizing a pattern
- Remembering what just happened
- Copying it
- Changing it
- Predicting what might come next
- Knowing what counts as a response
- Adjusting based on what the other person does

You clap/tap/make a sound → computer listens → responds with another pattern → you respond → it responds again. Then test whether it's actually responding to you or just generating random shit.

I hear what you did, compare it to a rule, and change it.

VOICE → PITCH DETECTION → COMPARISON → CORRECTION → MODIFIED VOICE

A person sings a note → software hears the pitch → decides where the note is supposed to be → changes it.

your voice → computer listens → detects pitch → compares it → changes it → corrected voice

The software has to recognize something about the sound before it can change it.

If I'm supposed to sing C but I'm slightly off, it basically goes:

"You're close to C → move this sound to C."

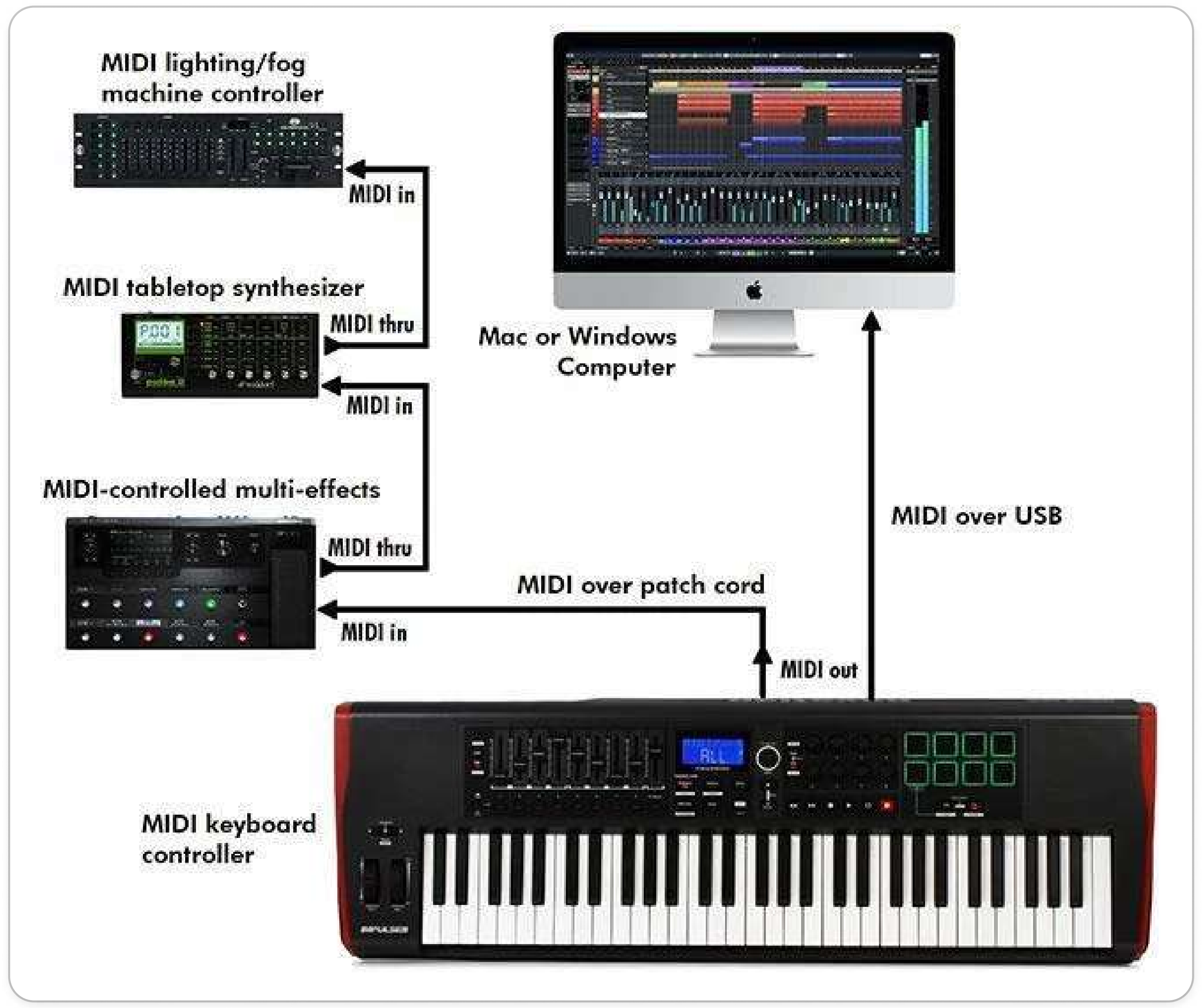
Does recognizing and correcting something mean you understand it?

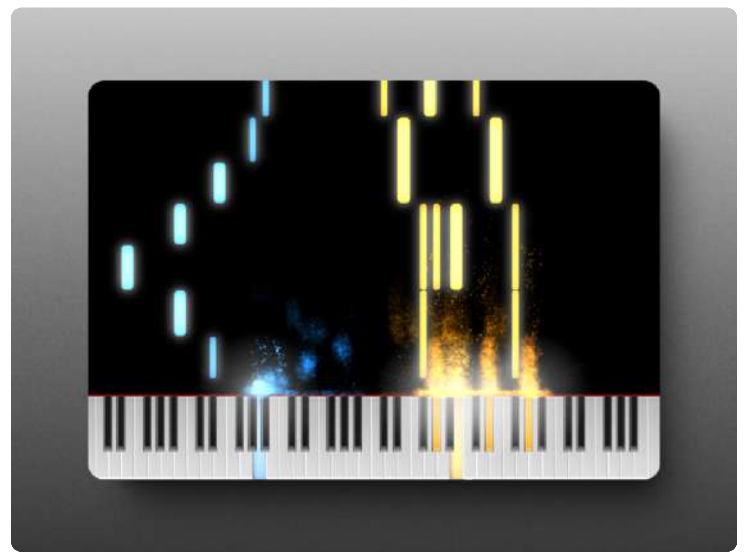
Where is the intelligence?
- How does it know what's "wrong"?
- Who decided what "correct" means?
- Is recognizing a pitch intelligence?
- Is correcting something intelligence?
- What happens when I intentionally sing between notes?
- What happens with talking, screaming, laughing, humming?
- Does it understand my voice, or is it literally just measuring it?

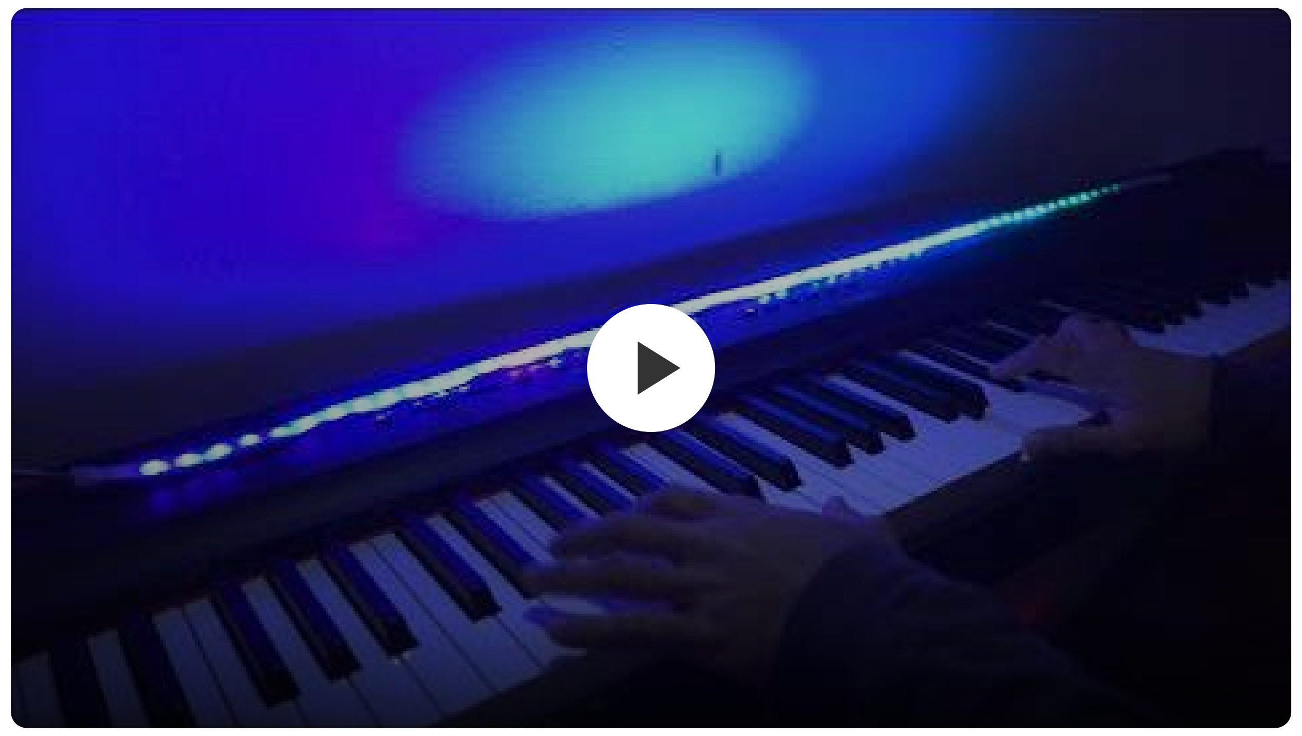
Give the software/code different vocal sounds and have it classify/change them according to a simple rule.
Then deliberately give it weird inputs and see where it fails.

Section 3

MIDI: FROM REPRESENTATION → RESPONSE







MIDI Lighting - Messing around with a Piano + LED...

YouTube

MIDI IS NOT THE SOUND

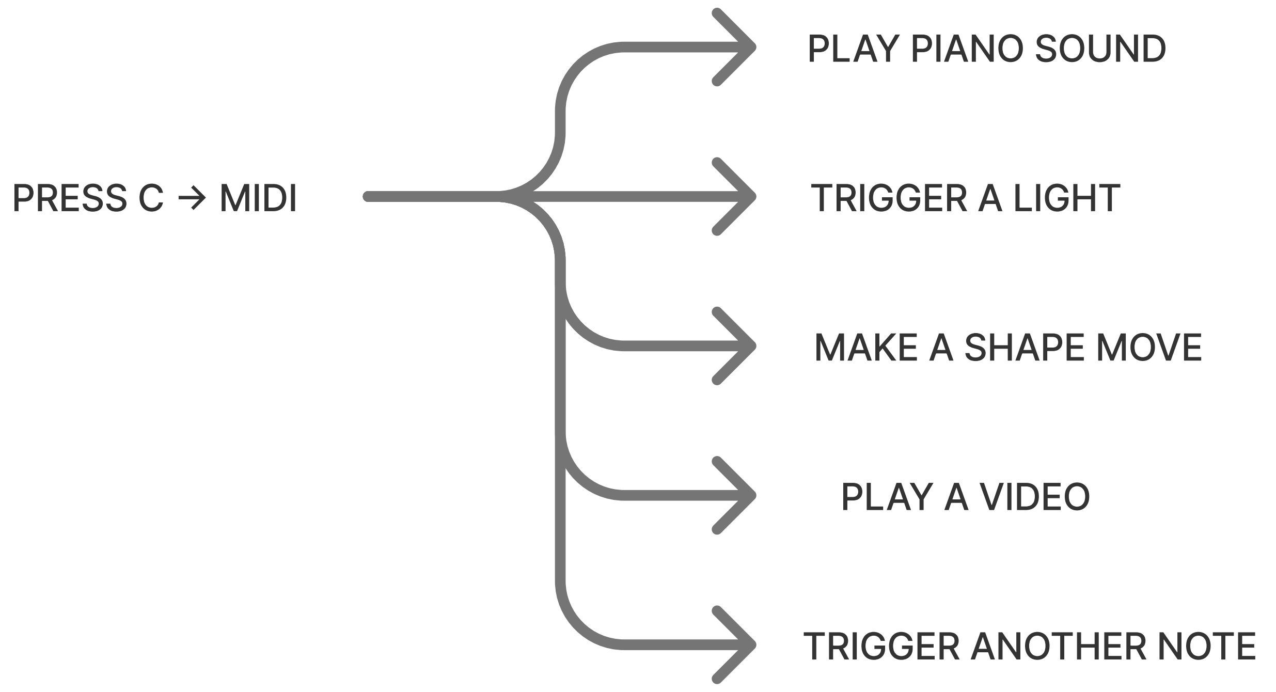
It carries instructions like:

WHAT NOTE?
WHEN?
HOW HARD?
HOW LONG?

ACTION → MIDI
INFORMATION → RESPONSE

The same MIDI information can cause completely different things to happen.

SAME INPUT → DIFFERENT RESPONSE



PRESS C → MIDI

- PLAY PIANO SOUND
- TRIGGER A LIGHT
- MAKE A SHAPE MOVE
- PLAY A VIDEO
- TRIGGER ANOTHER NOTE

The input stays the same.
The programmed rule
changes what happens next.

WHERE IS THE INTELLIGENCE?

IN THE PERSON PLAYING?

IN THE CODE?

IN THE MIDI INFORMATION?

IN THE RULES?

IN THE RESPONSE?

DOES THE SYSTEM UNDERSTAND ANYTHING?

IS IT JUST FOLLOWING INSTRUCTIONS?

BASIC SYSTEM

USER → MIDI → CODE → RESPONSE

INPUT
note / timing /
velocity / duration

RULE
IF this happens →
THEN do this

OUTPUT
sound / visual /
light / movement

C → THEN CIRCLE APPEARS
LOUDER → THEN CIRCLE GETS BIGGER
HELD LONGER → THEN CIRCLE LASTS LONGER

This isn't necessarily intelligent yet. It's just following rules.